

Meher Arush Burugula

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EDUCATION

University of Alberta

B.Sc. Computer Science and Mathematics

Calgary, AB

Sept 2024 – Present

EXPERIENCE

Utilyze

Backend Software Developer Intern

Calgary, AB

May 2026 – Present

- Develop backend data pipelines that research, aggregate, and process multiple utility outage datasets, transforming raw records into geospatial points rendered on a live interactive outage map.
- Work with machine learning models to analyze outage data and support predictive insight generation across mapped regions.

Cyberly (EVU Ventures)

Frontend Developer Intern

Remote

June 2026 – Present

- Build frontend features for a per-minute paid live video consultation marketplace connecting clients with professional experts, in a startup venture backed by EVU.
- Collaborate with a multi-disciplinary team of engineers and UI/UX designers on a platform integrating WebRTC real-time video and Stripe Connect payments.

Mangyanese Communications (Fruits of Change Foundation)

Volunteer Web Developer

Calgary, AB

May 2026

- Designed and built an 8-page e-commerce site from scratch; integrated Stripe payments and deployed to production via HostGator with a custom domain.

Maaco Autobody Shop

Parts Coordinator & Administrator

Calgary, AB

June 2025 – Sept 2025

- Processed repair estimates, sourced supplier parts, managed inventory, and handled customer scheduling and payments.

604 Moose Royal Canadian Air Cadets

Sergeant

Calgary, AB

Sept 2019 – Jan 2025

- Led and mentored junior cadets, delivered classroom presentations, and represented the squadron at public fundraising events.

PROJECTS

Vestr | Python, FastAPI, React, PostgreSQL, scikit-learn, yFinance, Railway, Cloudflare | [Live](#)

- Built and deployed a full-stack quantitative investment intelligence platform with live stock data, ML-powered buy/sell verdicts, and VADER sentiment analysis on real-time news headlines.
- Trained a Random Forest classifier on 5 years of OHLCV data across 100+ S&P 500 tickers (122,000+ records); achieved 47% accuracy on a 3-class prediction task, beating a 33% random baseline.
- Engineered a FastAPI backend with JWT auth, PostgreSQL, real-time price polling, a multi-stock screener with parallel threading, and price alerts; deployed via Cloudflare Pages with auto CI/CD.

Autonomous AI Agent | Python, Anthropic API, React, subprocess sandboxing | [GitHub](#)

- Built an autonomous agent that completes multi-step tasks end-to-end via a ReAct reasoning loop: the LLM selects tools, observes results, and iterates without human input at each step.
- Implemented a modular tool system (web search, file I/O, AST-validated calculator) with a defence-in-depth Python sandbox: subprocess isolation, OS-level resource limits, timeouts, and jailed file access.
- Developed a React frontend streaming the agent's reasoning and tool calls step-by-step in real time.

Digit Brain | Python, NumPy, Matplotlib | [GitHub](#)

- Built a 3-layer neural network from scratch in pure NumPy — no ML frameworks — implementing forward propagation, backpropagation, and gradient descent from first principles.
- Trained on MNIST (70,000 images), achieving 95%+ test accuracy with a live interactive canvas demo.

TECHNICAL SKILLS

Languages: Python, Go, JavaScript, TypeScript, HTML/CSS, SQL, Node.js

Frameworks: React, FastAPI, Express.js

Data & ML: scikit-learn, NumPy, pandas, yFinance, VADER, PostgreSQL, SQLite, Redis

Cloud & Tools: Git, Railway, Cloudflare, Vercel, Supabase, Azure, Oracle Cloud, CI/CD

Concepts: REST APIs, JWT Auth, LLM Tool Calling / Agents, Sandboxed Execution, ML Model Training, Neural Networks, Agile/SDLC

CERTIFICATIONS

Certified ScrumMaster – ScrumAlliance | Azure AI Fundamentals – Microsoft | Oracle Cloud Infrastructure Foundations – Oracle | Python Certification – Python Institute | Introduction to Generative AI – Google | Agile Foundations – LinkedIn Learning